MCAST

RESEARCH PROPOSAL FORM

(also referred to as the 'Statement of Intent Form', or SOI)

To be submitted by the researcher to the Institute Research Sub-Committee (IRC)

Research Title: Wearable In-Shoe System for Gait Anomaly Detection using FSR and IMU Sensors	
Institute name Institute of Information and communication Technology	
Course / Programme: Bachelor of Science (Honours) in Software Development	
Level and year of study Level 6 Year 3	
Main area of study being proposed: This project explores the design and development of a smart wearable system capable of continuously monitoring gait patterns and detecting anomalies using embedded sensors. The core objective is to identify abnormal gait behaviors, irregular motion, or excessive plantar pressure, all of which may indicate early biomechanical or health-related issues. Using an ESP32 microcontroller integrated with IMU (Inertial Measurement Unit) and FSR (Force Sensitive Resistors) sensors, the study will develop a cost-effective, wireless insole device for real-time gait anomaly detection. While the system has strong potential applications in diabetic foot monitoring, the research is designed to remain open and adaptable for broader contexts including rehabilitation, fall prevention in the elderly, and sports performance assessment.	
Name of Researcher: Joseph Daniel Buhagiar	**Researcher's I.D. Number:** 0179599M
Signature of Researcher	**Date of submission of Form**

JDBuhagiar	10/11/2025
Name of Tutor (or Recommended Tutor): Dr Clifford DeRaffaele	

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Personal Motivation for the Choice of Research Theme.

This research is motivated by personal experiences with the impact of diabetes on loved ones. My grandfather suffered from complications that led to multiple amputations before his passing, highlighting the severe consequences of delayed intervention. Additionally, my partner lives with type 1 diabetes, further reinforcing awareness of the daily challenges faced by diabetic individuals.
These experiences have inspired me to contribute to the development of preventative healthcare and mobility-monitoring solutions through technology. Although this study focuses primarily on gait anomaly detection, the diabetic context serves as a personal and meaningful motivation for exploring wearable sensor technology that can ultimately support early detection and prevention of foot-related complications.

Outline of Key Literature and Theoretical Framework or Propositions.

Introduction
Diabetic foot ulcers (DFUs) are among the most severe complications of diabetes, affecting up to 25% of patients during their lifetime and often leading to amputation if not addressed early [1]. Advances in embedded systems and wearable technology have enabled the development of smart insole devices capable of monitoring physiological indicators such as plantar pressure, temperature, and gait abnormalities. These metrics are closely linked with early DFU onset, making them promising targets for preventative health interventions [2]– [4].

This review outlines the research areas underpinning the development of a wearable diabetic foot monitoring system. It focuses on time-series analysis of foot health data, gait characterisation using IMU and FSR sensors, anomaly detection techniques, and relevant performance metrics. The section closes by integrating clinical research supporting smart insole efficacy.

Time-Series Analysis and Signal Processing

Time-series modelling is vital for tracking subtle changes in gait and foot biomechanics over time, particularly as abnormalities often develop gradually. Traditional clinical assessments may miss early signs of ulcer risk. ARIMA models and intervention analysis can detect such gradual deviations through repeated observations at the individual level [1].
Velicer and Fava [1] demonstrated that single-subject time series models are highly effective for recognising pressure-related micro-patterns. Jose [2] extended this approach to address real-time signal

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stability, ensuring that sensor data – often affected by drift and noise – could be filtered reliably. These methods enable the development of personalised gait baselines, allowing deviations to be flagged as potentially ulcerogenic [2].

Gait Characterisation with IMU and FSR Sensors

Building upon signal analysis, low-cost gait monitoring can be achieved by integrating force-sensitive resistors (FSRs) and inertial measurement units (IMUs). IMUs capture angular velocity and acceleration across multiple axes, while FSRs measure plantar pressure. Together, these sensors provide a rich picture of foot motion and pressure dynamics when integrated into systems like the ESP32 microcontroller.
For instance, Manupibul et al. [3] achieved less than 8% error in gait phase detection using an ESP32-based system with GY-85 IMUs and FSRs. Similarly, Gujarathi and Bhole [4] validated the use of the MPU6050 IMU for detecting gait phases such as heel-strike and toe-off. This phase information is crucial for assessing load distribution and identifying patterns indicative of diabetic risk.
Moreover, Ghazali et al. [5] introduced a Bluetooth-enabled insole integrating LM35 temperature sensors and multiple FSRs. Their system transmitted real-time pressure and temperature data to a smartphone interface, showing the potential of multimodal wearable systems to enable early detection of DFU risk. Collectively, these studies confirm the reliability and feasibility of sensor fusion in wearable diabetic foot monitoring.

Anomaly Detection Techniques

Rather than relying on general classification schemes, personalised health monitoring requires anomaly detection strategies tailored to individual users. One promising method is the Generalised One Class Anomaly Detection (GOAD) framework, which trains solely on ‘normal’ patterns and flags deviations through contrastive learning [6]. GOAD offers advantages over conventional autoencoders or PCA(Principal Component Analysis) based approaches, which tend to be sensitive to irregular data and noise. GOAD is especially effective in wearable applications where labelled examples of DFUs are scarce due to ethical and practical constraints. It can be learned from healthy gait patterns and flag outliers, potentially indicative of early pathology.
In practice, simpler models such as Z-score thresholds and Random Forest classifiers are also effective for real-time detection in resource constrained environments. These models offer a balance between sensitivity, computational efficiency, and interpretability key for embedded deployment [3], [5].

Evaluation Metrics for Gait-Based Systems

In the absence of confirmed DFU cases, surrogate indicators such as cadence, stance/swing ratios, pressure-time integrals, and deviations from baseline patterns are commonly used for evaluation. Z-score-based thresholding of these metrics can successfully identify outlier steps. Additionally, classifier performance is measured using precision, recall, and F1-score to assess model accuracy in labelling anomalies [3].
A landmark clinical trial by Abbott et al. [7] demonstrated the real-world value of smart insoles. In this study, participants at high risk of DFUs received real-time alerts when plantar pressure exceeded safe thresholds. The intervention group saw a 71% reduction in ulcer recurrence, rising to 86% among high-compliance users. This shows the potential impact of continuous foot monitoring systems in reducing complications.
Interestingly, the same study also noted that many high-risk events occurred during static periods like sitting or standing. This insight challenges the focus on peak walking pressure and highlights the value of low-frequency sampling for capturing cumulative stress an approach well-suited to embedded systems.

Conclusion
The reviewed literature supports the technical feasibility and clinical relevance of a smart insole system incorporating IMU, FSR, and possibly temperature sensors for early detection of DFU risk. Prior work has demonstrated the effectiveness of sensor fusion and machine learning techniques for capturing gait anomalies. This study builds on these foundations by using a minimalistic, low-cost configuration and

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lightweight anomaly detection techniques suitable for real-world use.

Although similar initiatives are already underway in Malta, this project contributes by offering a simpler, more accessible solution for continuous gait monitoring. Future iterations may incorporate adaptive thresholding and additional IMUs to improve robustness and extend applicability to varied terrains and user populations.

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Significance of the Study.

This research addresses a critical gap in accessible and affordable wearable technologies for gait monitoring and anomaly detection. By leveraging low-cost embedded systems and sensor fusion, the study aims to develop a real-time monitoring insole that can identify irregular gait behavior, helping prevent potential mobility issues and improving early intervention strategies. While the system has clear applications in diabetic care, it is also relevant to broader areas such as rehabilitation, elderly mobility monitoring, and sports performance optimization.

Hypotheses and/or Research Question/s

Hypothesis:
If a wearable smart insole can continuously monitor plantar pressure and gait patterns and detect deviations from personal baselines, then early-stage gait anomalies can be flagged through anomaly detection, supporting early interventions in various health and mobility contexts.

Research Questions:
- Can a single IMU and two FSR sensors reliably detect key gait events (heel-strike, toe-off, stance, swing)?
- What signal processing methods are suitable for extracting usable gait features in real time?
- How can detected anomalies in pressure and motion be mapped to potential early indicators of gait irregularity or risk?

Target Participants and Research Methods for Data Collection and Analysis

The initial phase of this study will involve a single healthy adult participant replicating normal gait patterns and validate the technical functionality of the wearable prototype.This phase will optimise hardware configuration, sensor placement, and data acquisition pipelines in a controlled setting. The prototype will consist of an ESP32 microcontroller, a single MPU-9250 Inertial Measurement Unit (IMU), and two Force Sensitive Resistors (FSRs) embedded within a wearable insole. The IMU will capture multi-axis motion

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data, while the FSRs will monitor plantar pressure at the heel and forefoot regions. Sensor data will be transmitted in real time to a personal computer via USB. The collected data will be processed using low-pass and complementary filters to reduce noise and estimate foot orientation. Gait cycles will be segmented using signal thresholding and feature extraction methods to identify key gait events such as heel-strike, toe off, and stance-to-swing transitions. Parameters such as cadence, step time, and stance/swing ratio will be calculated and compared with clinically established norms. Video footage or manually logged timestamps will be used to validate event detection accuracy. This study adopts a quantitative approach. Sensor data will be analysed using time-series filtering, feature extraction, and statistical modelling. Gait features such as cadence, step time, and stance/swing ratio will be computed. Anomalies will be flagged using both Z-score analysis and a supervised Random Forest classifier. While this phase involves healthy participants, future iterations may extend to specific populations such as diabetic individuals or those in rehabilitation to assess the system's broader applicability.

Anticipated Contributions of the Study.

This study contributes to the development of low-cost, scalable wearable technology for real-time gait anomaly detection. It bridges the gap between laboratory-based gait analysis and real-world usability by focusing on lightweight, embedded implementation. Although existing projects (such as the Smart Insole Technology for Diabetic Foot Management developed locally) address specific medical contexts, this research complements them by focusing on general gait-based anomaly detection adaptable to multiple use cases

Dissertation Project Plan.

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WP 2025/2026	Description	Sep	Oct	Nov	Dec	Jan	Fab	Mar	Apr	May	Jun	
1	**Idea formulation and Proposal**											
	1.1 Discussion with Supervisor											
	1.2 Mentor Allocation											
	1.3 SOI Submission											
2	Literature Review											
	2.1 Background Research											
	2.2 Sensor Integration Studies											
	2.3 Evaluation Metrics Review											
3	**Research Methodology**											
	3.1 Hypotheses & Questions											
	3.2 Pipeline Design											
	3.3 Data, Approach, Evaluation											
4	**Development**											
	4.1 Prototype Assembly											
	4.2 Internal Testing											
	4.3 System Refinement											
	4.4 Testing & Evaluation											
5	**Deliverables**											
	5.1 Dissertation Write-up											
	5.2 VIVA Preparation											
	5.3 Paper											

Ethical Considerations.

Refer to *guidance points below. You are also additionally required to read MCAST Document 074 'Research Ethics Policy and Procedure' that is available on the College website*

Research shall be conducted in such a manner so as to avoid any psychological and physical harm to humans and animals and financial damage to organizations:

1. *Only the supervisor and examiners will have access to any data gathered.*
2. *Participants will remain free to withdraw from the study at any time without having to provide any reason. In the case of withdrawal, all the records and information collection will be deleted.*
3. *The participant, who is the sole proprietor of the data provided, is granting that such data would be processed for this study purposes only.*
4. *The data collection process will be a transparent process.*
5. *All transcriptions and/or electronic recordings reflecting the data collected, once exhausted, are to be deleted*
6. *Confidentiality, anonymity and data protection procedures are to be ethically abided by.*
7. *The researcher would provide a soft copy of the study to the participant, if required.*

Enter details here regarding possibility of issues regarding confidential personal data:

The study will collect limited demographic information (age group and gender) alongside sensor data (e.g., gait metrics from IMU and FSR sensors). No names, contact information, or other personally identifiable data will be collected.

All data will be anonymised at the point of collection. Age and gender will be used solely for basic grouping and not linked to individual identities. Data will be stored securely on local storage. Participants will be informed of their rights and may withdraw from the study at any time without consequence.

Enter details here regarding possibility of physical harm:

The research involves minimal physical risk, as walking trials will take place on flat, non-slip, indoor surfaces. Participants will be instructed to wear their regular, comfortable footwear to avoid discomfort. No high-risk tasks or movements are involved. Personal Protective Equipment (PPE) is not required given the non-invasive and low-risk nature of the procedure.

Enter details here regarding possibility of moral harm:

All participants will be treated respectfully and sensitively throughout the research. Language and behaviour will be inclusive, avoiding any psychological, cultural, or spiritual offence. Participation is entirely voluntary, and informed consent will be obtained in advance. The study does not target or discriminate against any specific group.

Enter details here regarding possibility of business harm:

There are no anticipated business or commercial harms arising from this research. The study does not involve access to, collection of, or use of any proprietary business information, trade secrets, commercial contracts, pricing data, or other commercially sensitive material. Data collected are limited to anonymised sensor outputs (IMU readings, FSR pressure values) and minimal non-identifying demographic information (age group and gender). No company names, workplace information, or commercial activities will be recorded.

The prototype and associated software will be developed using publicly available hardware components (ESP32, MPU-9250/MPU-6050, FSRs) and open-source or self-authored code. Where third-party libraries are used, they will be properly attributed and licensed code will be respected; no reverse-engineering or extraction of proprietary software from commercial devices will take place.

To further minimise any theoretical risk of harm to businesses, the following measures will be implemented:

All data will be anonymised at source and stored locally on secure devices under the researcher's control; no corporate servers will be accessed or used.

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Data will not be published in a form that could be traced back to any commercial entity or reveal commercial practices. Published results will contain only aggregated, de-identified findings and technical descriptions.

Participants will be instructed not to use workplace or company-owned footwear/equipment during trials; testing will take place in neutral, non-commercial environments (e.g., indoor laboratory or private spaces).

No marketing, commercial endorsements, or partnerships are intended or required for the completion of the study.

In the unlikely event that a company or third party expresses concern about any aspect of the research, the researcher will cease the relevant activity and consult the supervisor and the Institute Research Sub-Committee to resolve the issue.

Given these safeguards and the non-commercial nature of the study, the risk of causing reputational or financial harm to any business is negligible. The primary aim is technical validation and scientific knowledge generation, with potential beneficial applications for clinical practice and public health rather than commercial advantage at the expense of existing businesses.

*Please see **Annex 1** for a sample Participant Information Letter and **Annex 2** for a sample Participant Consent Form. Student is to submit a copy of the proposed Participant Information Letter and Participant Consent Form where applicable. Both documents should be attached to the end of the SOI that is being submitted by the student.*

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List of Key References:

[1] W. F. Velicer and J. L. Fava, "Time Series Analysis," ResearchGate, Apr. 2003. [Online]. Available: https://www.researchgate.net/publication/229633091_Time_Series_Analysis

[2] J. Jose, "Introduction to Time Series Analysis and Its Applications," ResearchGate, Aug. 2022. [Online]. Available: https://www.researchgate.net/publication/362389180_INTRODUCTION_TO_TIME_SERIES_ANALYSIS_AND_ITS_APPLICATIONS

[3] U. Manupibul, R. Tanthuwapathom, W. Jarumethitanont, P. Kaimuk, W. Limroongreungrat, and W. Charoensuk, "Integration of force and IMU sensors for developing low-cost portable gait measurement system in lower extremities," *Sci. Rep.*, vol. 13, no. 1, p. 10653, Jun. 2023, doi: 10.1038/s41598-023-37761-2.

[4] T. Gujarathi and K. Bhole, "Gait analysis using IMU sensor," in *Proc. 10th Int. Conf. Comput., Commun. Netw. Technol. (ICCCNT)*, Jul. 2019, doi: 10.1109/ICCCNT45670.2019.8944545.

[5] A. Baihaqi, C. Y. Ming, and S. Mohamad, "Design of smart device for foot of diabetic patient in Malaysia," in *Proc. Int. Congr. Adv. Technol. Eng. (ICOTEN)*, Jul. 2021, pp. 1–6, doi: 10.1109/ICOTEN52080.2021.9493450.

[6] L. Bergman and Y. Hoshen, "Classification-based anomaly detection for general data," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2020. [Online]. Available: https://openreview.net/pdf?id=H1laeJrKDB

[7] C. A. Abbott et al., "Innovative intelligent insole system reduces diabetic foot ulcer recurrence at plantar sites: a prospective, randomised, proof-of-concept study," *Lancet Digit. Health*, vol. 1, pp. e308–e318, Oct. 2019, doi: 10.1016/S2589-7500(19)30153-2.

This section is to be filled in by the representative of the Institute Research Sub-Committee (IRC) prior to forwarding of this Form to the 'MCAST Research Ethics Committee' for final ethics approval:

Nature of Ethical Consideration	***Outcome (Tick)***	***Comments/Advice***
All ethical issues have been adequately tackled.		
Possibility of issues regarding misuse of data or some form of harm.		

Details of Representative to the Institute Research Sub-Committee.	
Name	Signature
Designation	Date

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Annex 1: Participant Information Letter

Sample:

Title of Research: _____

You are being invited to take part in a research study. Before you decide to participate, it is important for you to understand why the research is being done and what it will involve. Please take time to read the following information carefully and discuss it with others if you wish. Ask us if there is anything that is not clear or if you would like more information.

What is the purpose of the study?
This research is being undertaken on…

Why have I been chosen?
You have been chosen because…

Do I have to take part?
It is up to you to decide whether or not your take part. If you decide to take part you will be given this information sheet to keep and be asked to sign a corresponding consent form.

What will happen to me if I take part?
You will then be given a questionnaire on.../your data will be used…/your image will be used…

What are the possible disadvantages and risks of taking part?
There are no disadvantages or risks foreseen in taking part in the study.

What are the possible benefits of taking part?
By taking part you will be contributing to the development of a set of recommendations for…

What if something goes wrong?
If you wish to complain or have any concerns about any aspect of the way in which you have been approached or treated during the course of this study, please contact…(researcher is to give his/her MCAST email as a contact)

Will my details be kept confidential?
All information which is collected about you during the course of the research will be kept strictly confidential so that only the researcher carrying out the research will have access to such information and will not be shared with any other individuals. Participants should note that data/images collected from this project may be retained and published in an anonymized form. By agreeing to participate in this project, you are consenting to the retention and publication of data.

What will happen to the results of the research study?
The results will be written up into a dissertation for my final project of my Bachelor…

Who is organizing the research?
The research is conducted as part of a degree in …

Who may I contact for further information?
If you would like more information about the research before you decide…(researcher is to give his/her MCAST email as a contact)

Thank you for your interest in this research…

Annex 2: Participant (or Guardian) Consent Form

Sample:

Title of Research: _____

Name of Researcher: _____

Please initial box

1. I confirm that I have read and understand the Information Letter for the above study and have had the opportunity to ask questions. ☐
2. I understand that my/my charge's participation is voluntary and that I/my charge am/are free to withdraw at any time without giving any reason. ☐
3. I agree to allow my daughter/son/charge to take part in the above study. ☐

(Statement 3 is to be included only when guardians/parents are involved in giving consent)

_____	_____	_____
Name of Participant/ Guardian	Date	Signature
_____	_____	_____
Researcher	Date	Signature

1 for participant; 1 for researcher